

11 (short)

(a) Starting from the Biot-Savart law derive an expression for the magnetic flux density B on the axis of a circular coil of radius a , at a distance z from the plane of the coil, when the coil carries a current I . Sketch the result as a function of z for both positive and negative values of z . [5]

(b) Two such coils, carrying current I in the same direction, are placed a distance d apart on the same axis, as shown in Fig. 6. Sketch $B(z)$ as a function of z , along the axis of the coil. [3]

(c) When $d = a$, the magnetic flux density on the axis mid-way between the two coils has the property that $d^2B/dz^2 = 0$. What does this mean for the field around the midpoint? Comment on the usefulness of this result for experimental work. [2]

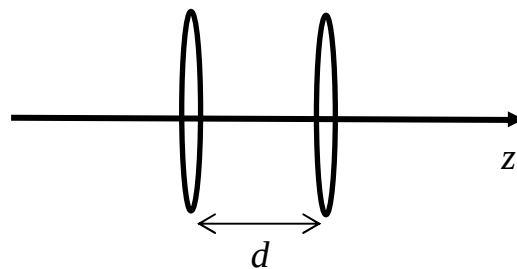


Fig. 6